

Course-Plan

Autumn Semester, 2026

School	ENGINEERING
Department	Computer Science and Engineering
Course Code	CSMT503
L-T-P-C	0-1-2-3
Course Name	Advanced Programming Lab I
Instructor	Shobhanjana Kalita

Pre-requisites: Data Structures

1. Abstract

This course will involve the students extensively in writing programs for assignments in contemporary programming languages. It covers programming of various advanced algorithms on matrix operations, linear programming, graph operations etc. The course further discusses implementation of advanced data-structures such as tries, height-balancing trees, and geometric data structures.

2. Course Objectives

- To develop competency in the use of procedural and object-oriented programming languages
- To enable students to design solutions using appropriate algorithmic techniques and data structures
- To develop competency in implementation of advanced algorithms
- To develop proficiency in implementation of advanced data structures

3. Course Outcomes

Students who complete the course will:

COs	Statements	Blooms Level
CO1	Understand the working of basic data structures	L2
CO2	Understand the working of advanced algorithmic methods	L2
CO3	Implement algorithms using procedural and object-oriented programming languages	L4
CO4	Apply advanced data structures for solving real world problems	L3

4. Course Plan

Topic	CO Mapping	No. of Classes
Basic Programming Skills (using C/C++/Java) 1. File handling 2. Basic data structures: linked list, matrix, stacks, queues 3. Basic algorithms: sorting, searching	CO1, CO3	4
Basic Algorithm Implementation 1. Recursions, divide and conquer 2. Greedy algorithms and dynamic programming	CO2, CO3	5
Graph Algorithms 1. Shortest Path algorithms 2. Minimum spanning trees 3. Network flow	CO1, CO3	5
Advanced data structures 1. Balanced trees, B-trees, B+-trees 2. Interval trees, skip lists, tries and suffix trees 3. Geometric data structures (R-tree, quadtree)	CO3, CO4	8
Matrix Operations 1. Inverse, Rank of a matrix 2. Eigen values and eigen vectors 3. Solving linear equations	CO3, CO4	8
Linear Programming	CO2	2

5. Laboratory experiments:

- i. Revision of file handling operations
- ii. Insertion, deletion, searching in linked lists
- iii. Merge-sort and quick sort using arrays
- iv. Recursion: Binary search
- v. Dynamic Programming for computing string similarity
- vi. Greedy problem solving for Interval Scheduling
- vii. BFS and DFS for searching in a graph
- viii. Prim's and Kruskal's algorithm for computing graph MST

- ix. Topological ordering of a DAG
- x. Matchings of a graph
- xi. Insertion, deletion, searching in AVL-tree/Red Black tree, B-tree, interval tree
- xii. Insertion, deletion, searching in R-tree, Quad-tree
- xiii. Multiplication, transpose, determinant, inverse of matrix
- xiv. Introduction to Cplex

6. Suggested Reading:

Textbook

- 1. Introduction to Scientific Computation and Programming, Daniel T. Kaplan
- 2. T. Cormen, C. Leiserson, R. Rivest, and C. Stein, Introduction to Algorithms, 2nd edition, 2001.

Supplementary Texts

- 1. Programming Pearls by Jon Bentley
- 2. Algorithms + Data Structures = Programs by Niklaus Wirth

6. Student Evaluation

Sl.no	Component	Marks
1	Test I, II, Midterm	10 + 10 + 30 = 50
2	End-Term	50

Total : 100

7. Pedagogy

Teaching-Learning Methods to be used:

- Tutorials and Discussion
- Quizzes and Programming Assignments